

Exam 1: Basics of R – Version B

Exam Agreement

This exam is an **individual assessment** and you should neither offer help to nor receive help from another person.

Cell phones are to be handed in to the instructor and all other electronic devices must be stored in bags at the front of the room.

No smart glasses or other wearable electronic devices are to be powered on during this exam except with accommodations.

You may use **no physical or electronic** resource to complete the work.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Exam Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: How Computers Work

“Hardware” often refers to the physical components of a computer, and “software” typically refers to programming. Explain how logic can be considered both software and hardware.

Question 2: Command line and syntax

The following line of code does not work:

```
df <- data.frame("words" = c("apple", "creature", "puppy"), "actions" = 1:3,  
                 nums = 4:7)
```

It produces the following error:

```
Error in data.frame(words = c("apple", "creature", "puppy"), actions = 1:3, :  
  arguments imply differing number of rows: 3, 4
```

Explain the source of the error. **Rewrite the line of code** that corrects the error and runs successfully below.

Question 3: Data Frames

In the `mtcars` data frame (Fig. 1), what is the mean miles per gallon (`mpg`) for 6-cylinder cars in the data set?

mtcars	32 obs. of 11 variables										
\$ mpg : num	21	21	22.8	21.4	18.7	18.1	14.3	24.4	22.8	19.2	...
\$ cyl : num	6	6	4	6	8	6	8	4	4	6	...
\$ disp: num	160	160	108	258	360	...					
\$ hp : num	110	110	93	110	175	105	245	62	95	123	...
\$ drat: num	3.9	3.9	3.85	3.08	3.15	2.76	3.21	3.69	3.92	3.92	...
\$ wt : num	2.62	2.88	2.32	3.21	3.44	...					
\$ qsec: num	16.5	17	18.6	19.4	17	...					
\$ vs : num	0	0	1	1	0	1	0	1	1	1	...
\$ am : num	1	1	1	0	0	0	0	0	0	0	...
\$ gear: num	4	4	4	3	3	3	3	4	4	4	...
\$ carb: num	4	4	1	1	2	1	4	2	2	4	...

Figure 1: The mtcars data set.

Question 4: Matrices and Arrays

You have a 3-dimensional array of dimensions 4, 4, and 8 named `array1`. How can you retrieve the third position of the first dimension, all of dimensions two, and the third position of the third dimension in a line of code?

Question 5: Lists

I have created a list stored with the name `my_list` with the following code:

```
my_list <- list(c("a", "b", "c", "d", "e"),
               matrix(runif(500), ncol = 50, nrow = 100),
               data.frame("x" = c(1.1, 1.2, 1.3),
                           "y" = c(2.3, 5.6, 6.2)),
               10:100)
```

- a. How many members does this list contain?
- b. What data class is each member?
- c. How would you reference the integer 15 in `my_list`?

Question 6: Class Coercion

Write the function that is capable of making the necessary conversion:

- a. a numeric vector into an integer vector:
- b. two vectors into a list:
- c. a data-frame column to a logical vector:
- d. a logical vector into an integer vector:

Question 7: Data Classes and Special Values

You are working with the `diamonds` data frame (Fig. 2), which is a data set of the cut, clarity, and price (among other values) of over 53,000 diamonds. Use code to complete the following questions:

▼ diamonds	53940 obs. of 10 variables	
\$ carat	: num [1:53940]	0.23 0.21 0.23 0.29 0.31 0...
\$ cut	: Ord.factor w/ 5 levels	"Fair"<"Good"<...: ...
\$ color	: Ord.factor w/ 7 levels	"D"<"E"<"F"<"G"<...: ...
\$ clarity	: Ord.factor w/ 8 levels	"I1"<"SI2"<"SI1"<...: ...
\$ depth	: num [1:53940]	61.5 59.8 56.9 62.4 63.3 62...
\$ table	: num [1:53940]	55 61 65 58 58 57 57 55 61 ...
\$ price	: int [1:53940]	326 326 327 334 335 336 336...
\$ x	: num [1:53940]	3.95 3.89 4.05 4.2 4.34 3.9...
\$ y	: num [1:53940]	3.98 3.84 4.07 4.23 4.35 3...
\$ z	: num [1:53940]	2.43 2.31 2.31 2.63 2.75 2...

Figure 2: The diamonds data set.

- You're concerned that the data set may be incomplete, but it's too large for you to scan. How can you check the entire data set to see if there are **any** missing values in it in one line of code that returns a single value?
- Assuming that there are missing values, how could you clean the data set so that only complete observations remain?

Question 8: Factors

Pick two columns from the `mtcars` data frame (Fig. 1) and convert each to factored data, either ordered or unordered. The two columns' data must make sense as a factor. Replace the original column with the new factored column.

Question 9: Vectors

Create a vector that contains 1,000 random numbers and take the mean of this vector in one line of code.

Question 10: Troubleshooting

You need to find a function that helps interface with git. How could you search the documentation to find this **with the console** (i.e., command line)?